

4.4.3 Inheritance

This topic builds on the study of cells in [Cells in animals and plants](#) (page 27) to explore the relationships from the molecular level upwards between genes, chromosomes and phenotypic features. Content covered includes sex determination in humans and single gene inheritance of particular characteristics. Included is the understanding that most phenotypic features are the result of multiple genes rather than single gene inheritance. The ideas presented here lead on to the study of mutations, selective breeding and genetic engineering in [Variation and evolution](#) (page 82).

4.4.3.1 Chromosomes and genes

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the following terms: gamete, chromosome and gene. Describe DNA as a polymer made up of two strands forming a double helix. Describe the genome as the entire genetic material of an organism.	Sexual reproduction involves the joining (fusion) of male and female gametes (sperm and egg cells in animals). In sexual reproduction there is mixing of genetic information, which leads to variety in the offspring. The formation of gametes involves meiosis. The genetic material in the nucleus of a cell is composed of a chemical called DNA contained in the chromosomes. Human body cells contain 23 pairs of chromosomes. DNA is made of very large molecules in long strands, twisted to form a double helix. A gene is a small section of DNA on a chromosome. Each gene contains the code for a particular combination of amino acids to make a specific protein. The genome of an organism is made up of all the genes in the DNA of its body cells.	This topic has links with Cells in animals and plants (page 27).

4.4.3.2 Sex determination in humans

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe sex determination in humans.	In human cells, one of the 23 pairs of chromosomes carries the genes that determine sex. In females the sex chromosomes are the same (XX); in males the chromosomes are different (XY). All eggs contain an X chromosome. Sperm cells contain either an X or a Y chromosome.	

4.4.3.3 Single gene inheritance

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain single gene inheritance. Predict the results of single gene crosses. Explain the terms allele/variant, dominant, recessive, homozygous, heterozygous.	Some characteristics are controlled by a single gene. Examples are fur colour in mice and red–green colour blindness in humans. Each gene may have different forms called alleles. A dominant allele is always expressed, even if only one copy is present. A recessive allele is only expressed if two copies are present (therefore no dominant allele present). If the two alleles present are the same the organism is homozygous for that trait, but if the alleles are different they are heterozygous.	WS 1.2 Complete a Punnett square diagram or interpret the results of a genetic cross diagram for a single gene, and understand family trees. MS 2e (HT only) Construct a Punnett square diagram to make predictions based on simple probability. MS 1c Use direct proportion and simple ratios in genetic crosses.

4.4.3.4 Genotype and phenotype

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe simply how the genome, and its interaction with the environment, influences the development of the phenotype of an organism. Explain the terms genotype and phenotype. Recall that most phenotypic features are the result of multiple genes rather than single gene inheritance.	All the genes present in an individual organism interact with the environment in which the organism grows and develops its observable appearance and character. These characteristics are its phenotype. The variation in the characteristics of individuals of the same kind may be due to differences in: • the genes they have inherited (genetic causes) • the conditions in which they have developed (environmental causes) • a combination of genes and the environment. Human height is an example of a characteristic determined by many genes, each with different alleles. The set of alleles that determine the height of a person is the genotype for that characteristic. Height is also affected by diet and exercise which are part of the environment in which an individual grows up.	WS 1.2 Explain why studies involving identical twins help to separate the contribution of genes and the environment to the development of their phenotypes. WS 1.1 Given a context and related information, discuss the potential importance for medicine of our increasing understanding of the human genome.